

PantryPal: Assignment 2

Information Retrieval (5LN712)
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1 Introduction

For this assignment I chose to create an AI PantryPal Agent. You can interact with the agent by telling it what you usually keep for food, what you like and dislike, and what you're hoping to make for dinner. Given the input query, the agent retrieves recipes it has in its persistent pantry memory, preference memory, a local recipe corpus, a substitution corpus, and a configurable web-search fallback

2 Architecture

PantryPal is organized around a small ReAct-style agent loop (1). The user interacts through a command-line interface. A rule-based planner classifies each message as pantry management, preference management, recipe search, substitution search, cooking technique lookup, or a broad cooking request. It then selects tools from a shared interface, where each tool accepts structured input and returns structured output. This keeps the system extensible: memory, retrieval, web search, and model access can be modified independently.

The architecture is intentionally retrieval-first. JSON-backed memory stores pantry ingredients and user preferences such as dietary restrictions, disliked ingredients, and favorite cuisines. Local recipes are loaded from `data/recipes`, while substitutions are loaded from `data/substitutions`. The agent composes tool observations into a final prompt for the language model. If no API key is available, the system still returns a grounded fallback answer containing retrieved recipe titles, substitutions, source labels, and a tool trace. This is important for reproducibility because the core assignment behavior can be demonstrated offline.

3 Reflection

For this assignment I again used a mix of Chat-GPT and Codex. I used Chat-GPT primarily to understand the high level idea of what we're creating and what different parts of the architecture were doing. I also used it to create efficient and small prompts to ensure readable and modular code. I had a lot of fun making the agent and hope to work on similar projects in my free time. I'm thinking of making some kind of perfume app, similar to Good Reads, where you can get list which perfumes you have, review them, and maybe even get suggestions of which perfumes you can mix together to achieve a nice smell (that's where API queries/AI comes in). Now that I know how to work more with agents, making that app seems possible :D

References

- [1] Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. *International Conference on Learning Representations*.